

Week 6 - Reading Week Consolidation: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

This is an optional consolidation sheet for Reading Week. It contains no new material and deliberately mixes ideas from Weeks 1–5.

YOU SHOULD TRY Question 1: A launch-day atmospheric state

At the surface, pressure is 1005 hPa and temperature is 285 K.

1. Calculate the air density using $R_d = 287 \text{ J kg}^{-1} \text{ K}^{-1}$.
2. Using a scale height of 8.2 km, estimate the pressure at 2.0 km.
3. Give one reason why the actual pressure at 2 km may differ from this simple estimate.

EXTRA CHALLENGE Question 2: Pressure as atmospheric mass

What fraction of the atmospheric mass lies above the 700 hPa pressure surface if mean sea-level pressure is 1013 hPa? What fraction lies below it?

YOU SHOULD TRY Question 3: From a pressure difference to a wind estimate

Across a 500 km distance, pressure decreases by 12 hPa. Take density $\rho = 1.10 \text{ kg m}^{-3}$ and latitude 50°N .

1. Calculate the horizontal acceleration due to the *pressure-gradient force alone*,

$$|a_p| = \frac{1}{\rho} \left| \frac{\Delta p}{\Delta x} \right|.$$

This is not the parcel's total horizontal acceleration: once the air is moving, Coriolis and other forces may also contribute.

2. Calculate $f = 2\Omega \sin \phi$ using $\Omega = 7.292 \times 10^{-5} \text{ s}^{-1}$.
3. Assuming straight isobars and negligible friction, estimate the geostrophic wind speed by balancing the pressure-gradient and Coriolis accelerations.

YOU SHOULD TRY Question 4: Put the dynamical chain in the correct order

A sunny coastline develops a sea breeze. The following statements are deliberately out of order:

- a near-surface pressure-gradient acceleration develops from sea toward land;
- the land warms faster than the sea;
- a closed circulation develops, with a return flow aloft;
- the air column over land becomes warmer and its pressure decreases more slowly with height;
- near-surface air begins to move onshore.

Put the statements into a physically sensible order. Then state whether Coriolis deflection would be more important for a weak local sea breeze lasting a few hours or for a synoptic-scale pressure system lasting several days, and explain why.

EXTRA CHALLENGE Question 5: Which approximation is being used?

For each statement, identify the approximation or physical law being invoked, and give one circumstance in which it could fail or need refinement.

1. $p = \rho R_d T$.
2. $dp/dz = -\rho g$.
3. $p = p_0 e^{-z/H}$ with constant H .
4. Upper-level wind is parallel to straight isobars.

YOU SHOULD TRY Question 6: Exam-style synthesis

A weather map shows tightly packed isobars over northern Britain and widely spaced isobars over southern Britain. A radiosonde launched in the north shows strong winds at 2 km.

Explain, using the pressure-gradient and Coriolis forces, why strong winds are expected in the north. Then explain why the near-surface wind direction and speed need not match the 2 km wind exactly.